

Bài 2: Code nối vào bài 1

```
28
29 # --- BÀI 2 ---
30 # GET /books/<id>
31 @app.get("/books/<int:bid>")
32 def fetch(bid):
33     i = next((k for k,b in enumerate(BOOKS) if b["id"] == bid), None)
34     if i is None: return jsonify(error="not found"), 404
35     resp = make_response(jsonify(BOOKS[i]), 200)
36     resp.headers["Cache-Control"] = "max-age=60"
37     return resp
38
39 # PUT - thay toàn bộ, title+author bắt buộc
40 @app.put("/books/<int:bid>")
41 def put(bid):
42     i = next((k for k,b in enumerate(BOOKS) if b["id"] == bid), None)
43     if i is None: return jsonify(error="not found"), 404
44     p = request.get_json(silent=True) or {}
45     t, a = p.get("title"), p.get("author")
46     if not t or not a: return jsonify(error="need title+author"), 422
47     BOOKS[i] = {"id":bid, "title":t.strip(), "author":a.strip(),
48 "isbn":p.get("isbn"), "price":p.get("price")}
49     return jsonify(BOOKS[i]), 200
50
51 # PATCH chỉ cập nhật field có trong body
52 @app.patch("/books/<int:bid>")
53 def patch(bid):
54     i = next((k for k,b in enumerate(BOOKS) if b["id"] == bid), None)
55     if i is None: return jsonify(error="not found"), 404
56     p = request.get_json(silent=True) or {}
57     if p.get("price", 0) < 0:
58         return jsonify(error="price must be positive"), 422
59     for k in "title author isbn price".split():
60         if k in p: BOOKS[i][k] = p[k]
61     return jsonify(BOOKS[i]), 200
62
63 # DELETE - idempotent, trả 204
64 @app.delete("/books/<int:bid>")
65 def delete(bid):
66     i = next((k for k,b in enumerate(BOOKS)
67 if b["id"] == bid), None)
68     if i is None: return jsonify(error="not found"), 404
69     BOOKS.pop(i); return "", 204
70
71 if __name__ == "__main__":
72     app.run()
```

1. POST để tạo trước 1 cuốn sách (Để có ID dùng cho việc test)

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/soa (main)
$ curl -i -X POST localhost:5000/books -H "Content-Type: application/json" -d '{"title":"Clean Code","author":"R. Martin"}'
HTTP/1.1 201 CREATED
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:02:31 GMT
Content-Type: application/json
Content-Length: 51
Location: /books/2
Connection: close

{"author":"R. Martin","id":2,"title":"Clean Code"}
```

2. GET id = 1 (max-age = 60)

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i localhost:5000/books/1
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:03:53 GMT
Content-Type: application/json
Content-Length: 51
Cache-Control: max-age=60
Connection: close

{"author":"R. Martin","id":1,"title":"Clean Code"}

```

3. PUT thay thế toàn bộ thông tin sách

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X PUT \
localhost:5000/books/1 \
-H "Content-Type: application/json" \
-d '{"title":"Clean Architecture","author":"Robert C. Martin","isbn":"978-0134494166","price":35.0}'
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:05:06 GMT
Content-Type: application/json
Content-Length: 103
Connection: close

{"author":"Robert C. Martin","id":1,"isbn":"978-0134494166","price":35.0,"title":"Clean Architecture"}

```

4. PATCH cập nhật một vài field (ở đây đổi price)

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X PATCH \
localhost:5000/books/1 \
-H "Content-Type: application/json" \
-d '{"price":19.99}'
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:06:08 GMT
Content-Type: application/json
Content-Length: 104
Connection: close

{"author":"Robert C. Martin","id":1,"isbn":"978-0134494166","price":19.99,"title":"Clean Architecture"}

```

5. DELETE

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -X DELETE -i \
localhost:5000/books/1
HTTP/1.1 204 NO CONTENT
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 15:06:44 GMT
Content-Type: text/html; charset=utf-8
Connection: close

```